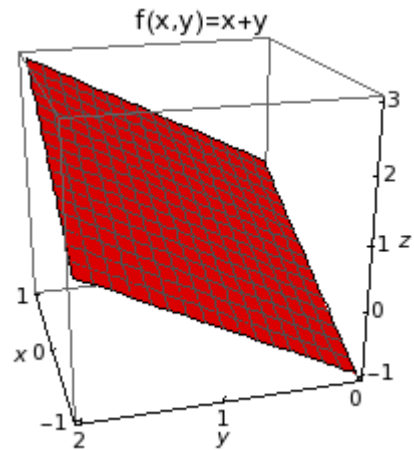


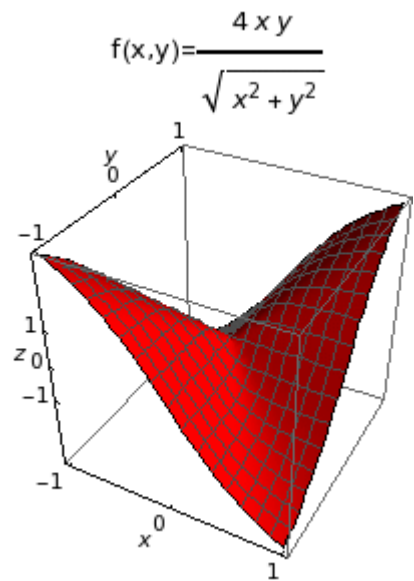
2

Exercício 2.1

a)

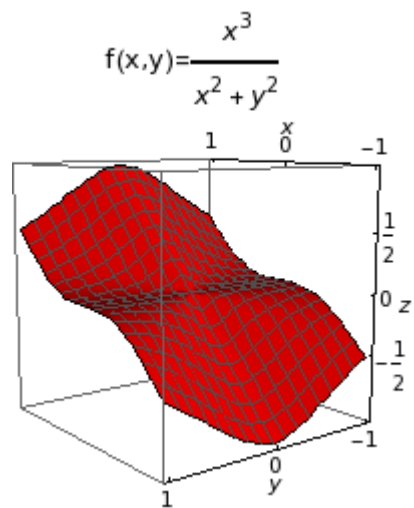


b)

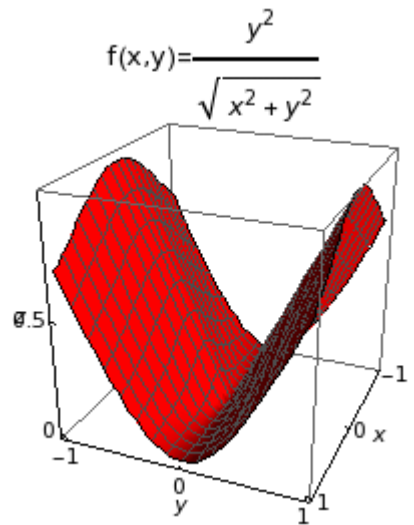


Exercício 2.2

a)

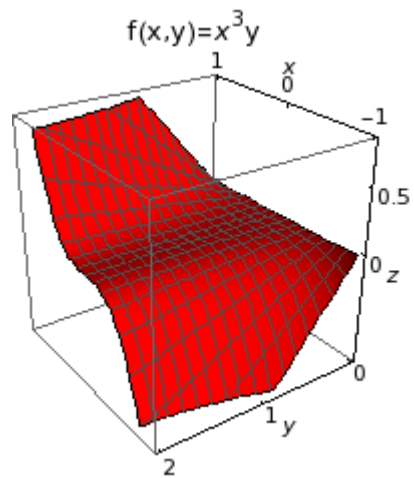


b)



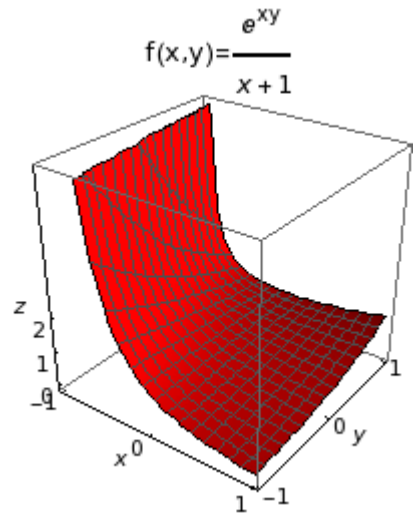
Exercício 2.3

a)



$$\lim_{(x,y) \rightarrow (0,1)} f(x,y) = 0$$

b)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$

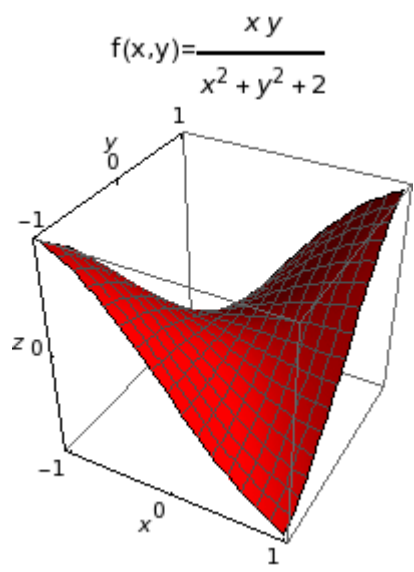
c)

$\lim_{x \rightarrow 1} (x^2, e^x) = (1, e)$

d)

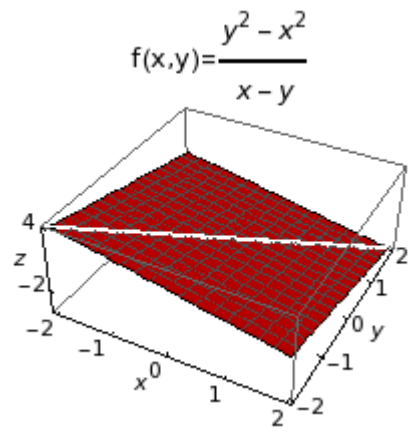
$\lim_{(x,y) \rightarrow (0,0)} \left(\frac{\cos [x]}{x^2 + y^2 + 1}, e^{x^2} \right) = (1, 1)$

e)



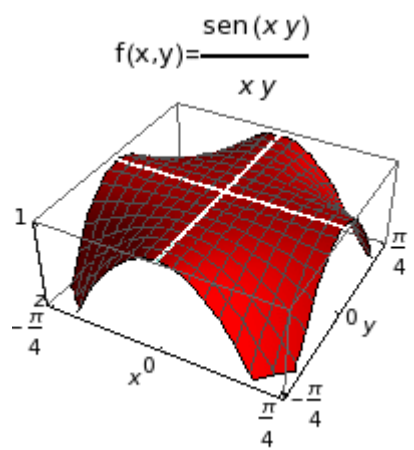
$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

f)



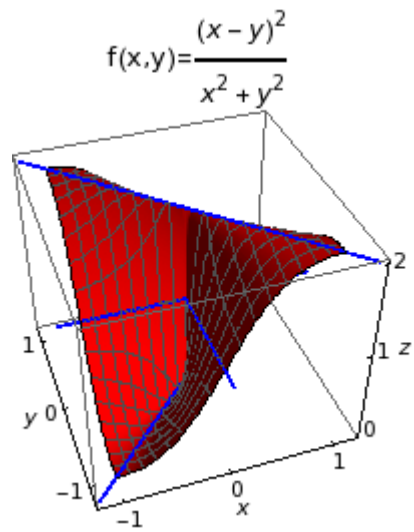
$\lim_{(x,y) \rightarrow (1,1)} f(x,y) = -2$

g)



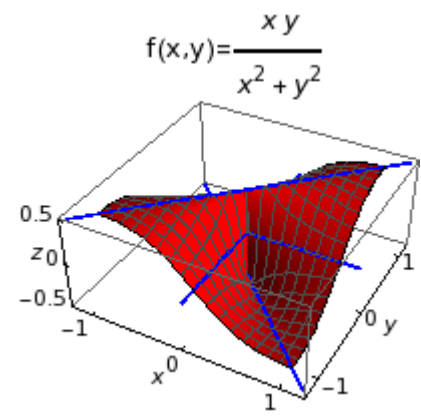
$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$

h)



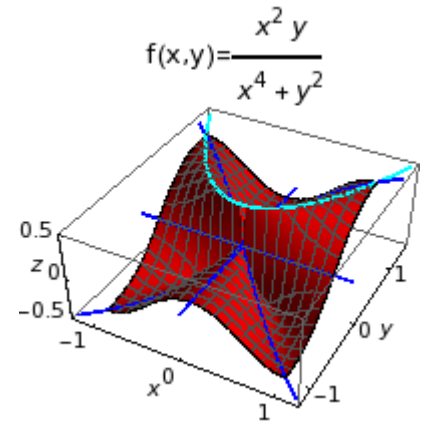
Não existe $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

i)



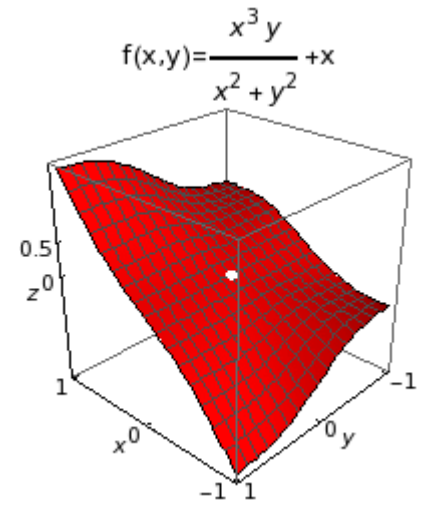
Não existe $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

j)



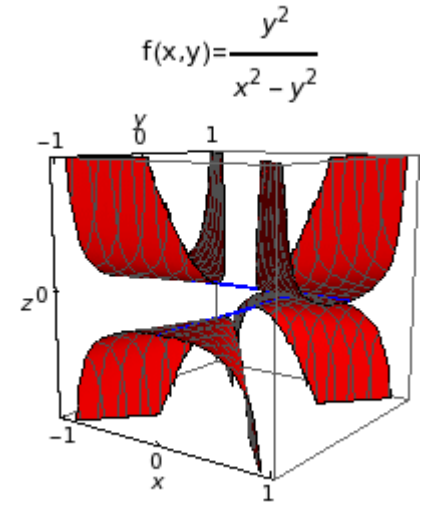
Não existe $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

k)



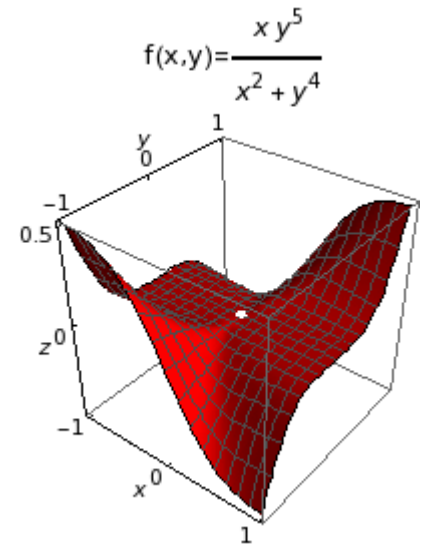
$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

l)



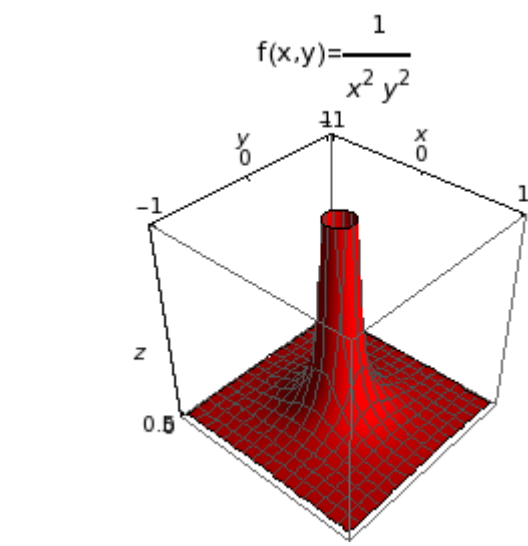
Não existe $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$

m)



$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$

n)



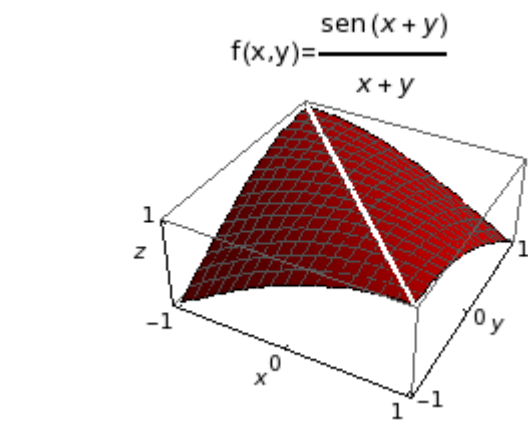
$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = +\infty$$

o)

$$\lim_{(x,y,z) \rightarrow (-1,1,0)} f(x,y,z) = (-1,0)$$

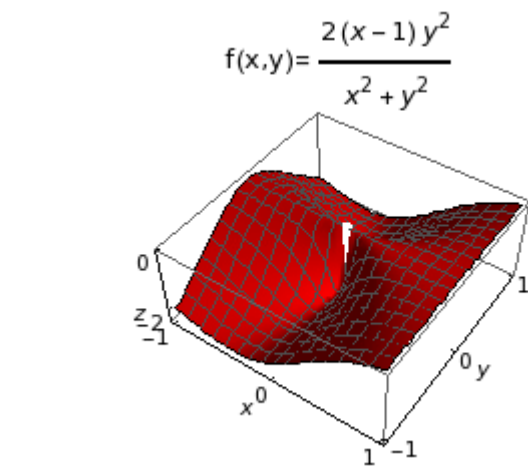
Exercício 2.4

a)



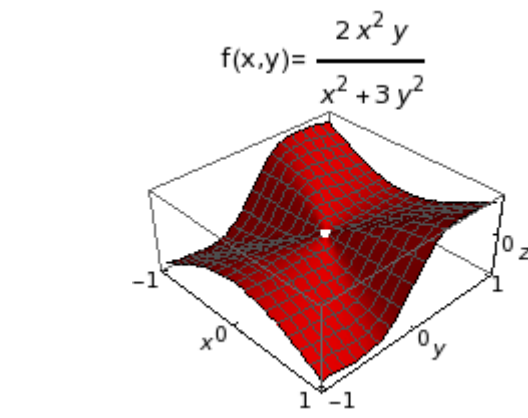
$$f(0,0) = 1$$

b)



Não admite prolongamento contínuo à origem

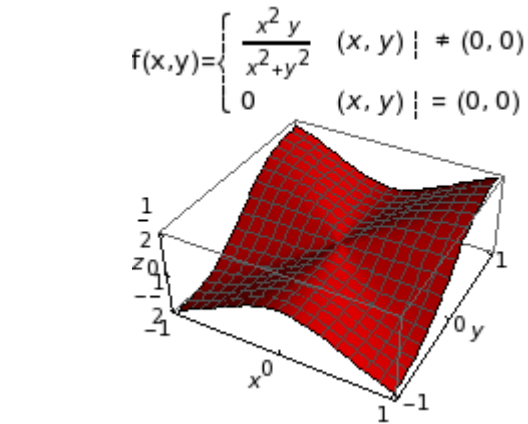
c)



$$f(0,0) = 0$$

Exercício 2.5

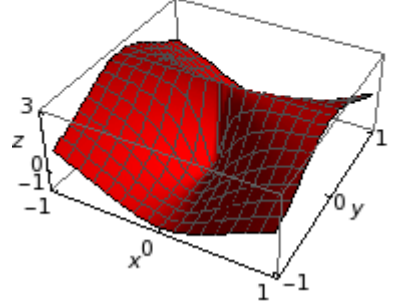
a)



A função é contínua em $\mathbb{R}^2$

b)

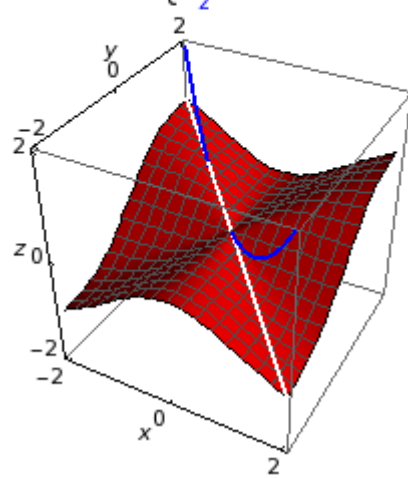
$$f(x,y)=\begin{cases} \frac{3x^2-y^2}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$



A função é contínua em $\mathbb{R}^2 \setminus \{(0, 0)\}$

c)

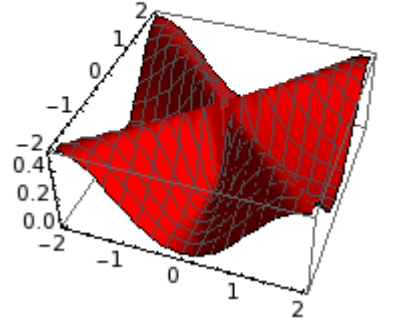
$$f(x,y)=\begin{cases} \frac{x^2 y}{x^2+y^2} & x \neq -y \\ \frac{x^2}{2} & x = -y \end{cases}$$



A função é contínua em $\{(x, y) \in \mathbb{R}^2 : x \neq -y\} \cup \{(0, 0), (-1, 1)\}$

d)

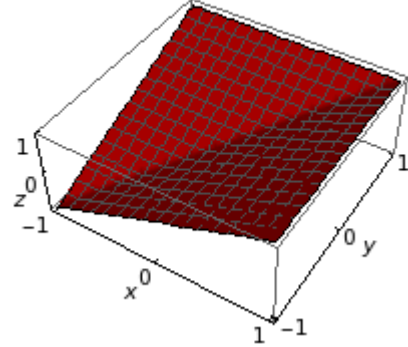
$$f(x,y)=\begin{cases} \frac{x^2 y^2}{x^4+y^4} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$



A função é contínua em $\mathbb{R}^2 \setminus \{(0, 0)\}$

e)

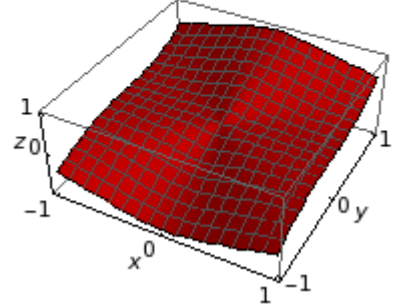
$$f(x,y)=\begin{cases} x & x \geq y \\ y & x < y \end{cases}$$



A função é contínua em $\mathbb{R}^2$

f)

$$f(x,y)=\begin{cases} \frac{y^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$



A função é contínua em $\mathbb{R}^2$

g)

A função é contínua em $\mathbb{R}^3$